MAHATMA GANDHI UNIVERSITY, KOTTAYAM
MGU_UGP (HONOURS) REGULAR/ IMPROVEMENT/REAPPEARANCE EXAMINATION NOVEMBER 2025
FIRST SEMESTER
Discipline Specific Core Course (DSC) – MG1DSCNBC100-
FUNDAMENTALS OF NUTRITIONAL BIOCHEMISTRY

Duration: 2 Hrs **Maximum marks: 50**

PART A
Answer all Questions
Each question carries 1 mark

1.B) Covalent bond
2.B) They must be obtained through diet
3.D) Helps in hormone regulation
4.B) Antioxidant defense
5. B) Collagen
6. B) Hydrogen Bond

Part B
Answer any two Questions
Each question carries 2 marks

7. Explain the physiological classification of carbohydrates

Simple carbohydrates (1 mark)
Complex carbohydrates (1 mark)

8. Describe the optical isomerism of amino acids and its dependence on pH and side chains.

When a beam of plane polarized light is passed through a solution exhibiting optical activity , it will be rotated to the right or left in accordance with the type of compound. (1 mark)
All amino acids except glycine have asymmetric carbon atom. (1 mark)

9. Summarize the classification of fatty acids based on the hydrocarbon chain

Short chain fatty acids: fewer than 6 carbon atoms. (1/2 mark)
Medium chain fatty acids: 6-12 carbon atoms. (1/2 mark)
Long chain fatty acids: 13-21 carbon atoms. (1/2 mark)
Very long chain fatty acids: more than 22 carbon atoms. (1/2 mark)

10. Describe the role of co-enzyme forms of niacin

NAD (Nicotinamide adenine dinucleotide) .(1 mark)
NADP (Nicotinamide adenine dinucleotide phosphate) involved in redox reactions.(1 mark)

Part C
Answer any 4 questions
Each question carries 5 marks

11. Explain the biological role of vitamin A in our body.
Vision
Immune system support
Cell growth and differentiation
Skin health
Antioxidant activity
Reproductive health
(any 5 points each carries 1 mark)

12. Explain the structure of fatty acids distinguishing between saturated and unsaturated fatty acids.

Fatty acid is an organic molecule made up of long hydrocarbon chain and a carboxyl group. (1 mark)
Saturated fatty acids: no double bonds
Eg. Stearic acid (1 mark)
Unsaturated fatty acids: one or more double bonds (1 mark)
Monounsaturated fatty acid: only one double bond. Eg. Oleic acid (1 mark)
Polyunsaturated fatty acid: more than one double bond. Eg. Linoleic acid (1 mark)

13. Discuss the role of phospholipids and cholesterol

Cholestrol:
Cell membrane structure
Production of vitamin D
Precurser for steroid hormones
Bile acid formation
Protection of nerve cells. (any four points ½ mark each)

Phospholipids:
Cell membrane structure
Electron transport chain component
Transport of lipids
Membrane fluidity
Signaling. (any four points ½ mark each)

14. Describe the structure and properties of riboflavin

Riboflavin is composed of an **isoalloxazine ring** attached to **a ribitol** side chain

Isoalloxazine ring is the core structure of riboflavin and consists of three fused rings: Benzene ring: A six membered ring with altering double bonds, containing carbon and hydrogens. Pyrazine ring: A six membered ring with nitrogen atoms at positions 1 and 4

Pyramidine ring: Another six membered ring containing nitrogen atoms at positions 1 and 3.

Attached to the isoalloxazine ring is a ribitol side chain (3marks)

Properties:

Water soluble vitamin

Heat stable

Destroyed by light

Yellow colour of urine (2 marks)

15. Discuss on the applications of radioactivity

Research field (2 marks)

Tracer technique

Isotope dilution technique

Carbon dating

Nucleic acid synthesis

Diagnosis of diseases (2 marks)

Life span of RBC

Thyroid scanning

Radioimmunoassay

Treatment of disease (1 mark)

Thyroid cancer treatment

Radium needles

Teletherapy

16. Discuss the classification of lipids

Simple lipids: (2 marks)

Fats and oils : esters of fatty acids with glycerol

Waxes: esters of fatty acids with alcohols other than glycerol

Compound lipids: (2mark)

Esters of fatty acids with alcohols containing additional groups.

Phospholipids, Glycolipids, lipoproteins

Derived lipids: (1mark)

these are the derivatives obtained from the hydrolysis of simple and compound lipids.

Part D
Answer any 2 questions
Each question carries 10 marks

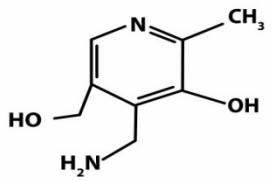
17. Discuss the properties of carbohydrates
Colour: colourless
Shape: Crystalline
Solubility: readily soluble in water
Taste: sweet in taste
Optical activity: D and L forms
Reducing sugar: All monosacharides and disaccharides (except sucrose) are reducing sugars
Methylation:Addition of methyl group
Glycosidic linkage: sugar molecules are joined by dehydration reaction
Dehydration reaction: removal of water molecule
Esterification: Glucose reacts with 5 molecules of acetic anhydride to form esters.
Mutarotation: optical rotation of α anomer and β anomer
(any 10 points 1 mark each)

18. Describe the properties and classification of proteins
Classification:
Based on source of protein: (2 marks)
Animal protein: derived from animal sources
Plant protein: derived from plant source.
Based on shape of protein molecule: (2 marks)
Fibrous protein: resemble long ribbons or fibresin shape. Eg. Keratin, collagen
Globular protein: possess a spherical or ovoid shape. Eg. Hemoglobin, myoglobin
Based on functions: (4 marks)
Defence proteins: defend or protect . eg. Antibodies
Contractile proteins: ability to contract. Eg. Actin, myosin
Transport proteins: transport of esstential factors. Eg. Hemoglobin, lipoproteins
Structural proteins: cell wall and structural component. Eg. Collagen, fibroin
Enzymic proteins: catalytic activity. Eg. Urease, amylase
Nutrient proteins : act as nutrient. Eg. Ovalbumin, casein
storage proteins: storage of nutrients. Eg. Ferritin
regulatory proteins: regulatory action. eg. insulin and growth hormone
toxic proteins: some proteins are toxic substances. Eg. Snake venom
On the basis of composition and solubility: (2 marks)
Simple proteins: yield only amino acids on hydrolysis. Eg. Protamines, histones, albumins
Conjugate proteins: In addition to amino acids, contain a non-protein group. Eg. Nucleoproteins, mucoproteins, chromoproteins, phosphoproteins, lipoproteins, metalloproteins.
Derived proteins: formed from simple and conjugate

Primary derived: heat, x-ray, acid , alkali. Eg. Proteans, metaproteins
Secondary derived: hydrolysis of proteins. Eg. Proteoses, peptones

19. Describe the structure and properties of pyridoxine

- Vitamin B6 is a water-soluble vitamin that exists in three main forms: pyridoxine, pyridoxal, and pyridoxamine, which are interconvertible.
- The chemical structure of each form includes a pyridine ring with different side groups. The chemical structure of pyridoxine,is 2-methyl-3-hydroxy-4,5-dihydroxymethylpyridine.
- The core structure of pyridoxine is a six-membered pyridine ring with nitrogen at one position (position 1).This ring has five carbons and one nitrogen atom, with alternating single and double bonds, giving it aromatic stability.
- It consists of Hydroxymethyl Group (-CH,OH): Attached to carbon 4 of the pyridine ring, then a Methyl Group (-CH3): Attached to carbon 2 of the ring. And hydroxyl Group (-OH): Attached to carbon 3 of the ring. Together, these groups give pyridoxine its distinctive structure and biochemical properties.
- The active coenzyme form of vitamin B6 is pyridoxal 5'-phosphate (PLP) (4 marks)
- Structure (2marks)



Properties: (4 marks)
• Vitamin B6 is water-soluble, and excess amounts are excreted in the urine.
• It is sensitive to heat and light, so food processing and cooking can cause some degradation

20. Explain the biological function of folic acid

- Vitamin B9, water soluble vitamin (1 mark)
- Folate is crucial for DNA synthesis, repair, and methylation, playing a key role in cell division and growth, especially in rapidly dividing tissues.
- It is involved in the synthesis of nucleotides (building blocks of DNA and RNA) and amino acids.
- It helps in the conversion of phenylalanine to tyrosine
- Synthesis of purine and pyramidine bases needed for the synthesis of DNA and RNA
- Folate helps in the formation of red and white blood cells in the bone marrow.
- It works closely with vitamin B12 in maintaining homocysteine levels, reducing the risk of cardiovascular diseases.
- It is especially important during pregnancy for fetal development, as it helps prevent neural tube defects. (7 marks)

Properties: (2 marks)
Folate is water-soluble, and excess amounts are excreted in urine.
I is sensitive to heat, light, and oxidation, meaning some folate can be lost during cooking and food processing.